

Yi-Hsin Tsai

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RESEARCH INTERESTS

My research interests broadly encompass complex geometry, microlocal analysis, and quantization theory. I am particularly interested in the asymptotic analysis of kernel functions such as Bergman and Szegő kernels, the theory of Toeplitz and Fourier integral operators, and their connections to geometric invariants and deformation quantization. My work aims to deepen the understanding of how analytic and symbolic techniques interact with geometric structures, including CR, contact, and conformal geometries, and to explore new frameworks that bridge analysis, geometry, and mathematical physics.

EDUCATION

Master of Science (M.S.) in Mathematics *Advisor: Chin-Yu Hsiao* June 2025

National Taiwan University

Bachelor of Science (B.S.) in Mathematics June 2023

National Taiwan University

ACADEMIC EXPERIENCE

NCTS | *Research Assistant* Aug 2025 - Present

NTU | *Project Research Scholarship Recipients* Oct 2023 - Jun 2025

- Bergman Kernel on Complex Manifold or Complex Orbifold.
- Advisor: Prof. Chin-Yu Hsiao

NCTS | *Research Abroad Program (at University of Cologne)* Aug 2024 - Nov 2024

SLMath | *Summer School* Jun 2024

- Special Geometric Structures and Analysis

NTU | *Teaching Assistant of Introduction of Geometry* Sep 2023 - Dec 2023

NCTS | *URP* Oct 2022 - Jun 2023

- Heat and Bergman Kernels Asymptotics in Complex Geometry.
- Advisor: Prof. Chin-Yu Hsiao

NTU | *Teaching Assistant of Complex Analysis* Sep 2022 - Dec 2022

NCTS | *USRP* Jul 2022 - Aug 2022

- Comparison of Various Methods in Deformation Quantization
- Advisor: Prof. Chin-Yu Hsiao

NTU | *Project Research Scholarship Recipients* Nov 2021 - Apr 2022

- Dynamics in One Complex Variable and some Fundamental of Complex Geometry.
- Advisor: Prof. Hsueh-Yung Lin

AWARDS/SCHOLARSHIP

Dean's Awards | *College of Science, NTU* 2025

Hu DunFu Memorial Scholarship | 2024

RESEARCH TALKS/PRESENTATIONS

Gateway to the World-Student Presentations after Academic Visits Abroad | *NCTS* Dec 2024

Stationary Phase Method on the Full expansion of Bergman Kernel

Seminar semiklassische Analysis und Darstellungstheorie | *University of Cologne* Nov 2024

Scaling Method of Bergman Kernel